Scrum Simulation Project: First Report (Group 7)

Smart Grocery Shopping List App

Group Members:

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Introduction

In the ever evolving digital world, traditional methods of grocery shopping are often flawed in terms of organization and efficiency. Forgotten items, excessive purchases, and food waste are common challenges faced by consumers. To address these issues, our smart grocery shopping list app has been designed to streamline and enhance the grocery shopping experience.

Our mobile application will help users to create and manage their shopping lists with ease. The key features include adding, editing, and categorizing grocery items, while also offering smart recommendations based on previous purchases. Additionally, expiration alerts for perishables and meal planning suggestions further simplify shopping decisions.

To improve efficiency, the app integrates barcode scanning for seamless pantry tracking and it also supports real time shared lists, enabling better collaboration among friends and family. By combining ai powered recommendations, location based reminders, and collaborative features, this app transforms the traditional grocery list into an intelligent shopping assistant that promotes smarter shopping habits, minimizes waste, and saves valuable time.

Overview of Scrum Process

To ensure a structured and effective development process, the scrum framework was implemented throughout sprint 1. This mobile app helps users organize grocery lists, track pantry inventory, and get smart shopping suggestions based on previous purchases and meal planning.

Product Backlog:

Product: Smart Grocery Shopping List App

1. Basic Grocery List Management

- "As a user, I want to add items to my grocery list so that I can remember what to buy."
- "As a user, I want to delete or edit items on my list so that I can update my shopping needs."
- "As a user, I want to categorize items (fruits, dairy, vegetables) so that I can organize my shopping."

2. Smart Suggestions & Reminders

- "As a user, I want to receive suggestions for frequently bought items so that I don’t forget to buy them."
- "As a user, I want to get expiry reminders for perishable items so that I can avoid food waste."
- "As a user, I want to receive notifications when I am near a grocery store so that I don’t miss shopping."

3. Pantry Inventory Tracking

- "As a user, I want to scan barcodes of purchased items so that my pantry inventory updates automatically."
- "As a user, I want to check what I already have in my pantry so that I don’t buy unnecessary items."
- "As a user, I want to set expiration dates for perishable items so that I can track freshness."

4. Meal Planning & Recipe Integration

- "As a user, I want to get meal plan recommendations based on my grocery list so that I can plan meals easily."
- "As a user, I want to see recipes based on available ingredients so that I can cook efficiently."
- "As a user, I want to add missing recipe ingredients to my grocery list so that I can buy them."

5. Multi-User Collaboration

- "As a user, I want to share my grocery list with family members so that we can coordinate shopping."
- "As a user, I want to receive updates when someone buys an item so that I don't purchase it again."

Sprint Planning:

The team established a clear sprint goal which is to develop a functional grocery list with core tracking and management features. The sprint backlog was created by prioritizing key user stories such as:

- Adding, editing, and deleting grocery list items.
- Categorizing items for better organization.
- Ensuring a simple ui/ux design for improved usability.
- Implementing basic data storage for saving and retrieving user data efficiently.

Sprint Review Outcome:

- Basic grocery list functionality works.
- Users can add, edit, and categorize items.
- Next sprint will add smart suggestions and pantry tracking.

Development Work:

During the sprint, the team focused on building core functionality first, ensuring the fundamental features aligned with the sprint goal.

Daily Standups:

The team conducted regular standup meetings to discuss progress, address blockers, and maintain alignment with the sprint goal. These meetings played a crucial role in identifying challenges early and keeping the development process on track.

Sprint Backlog:

❖ **Sprint Goal**

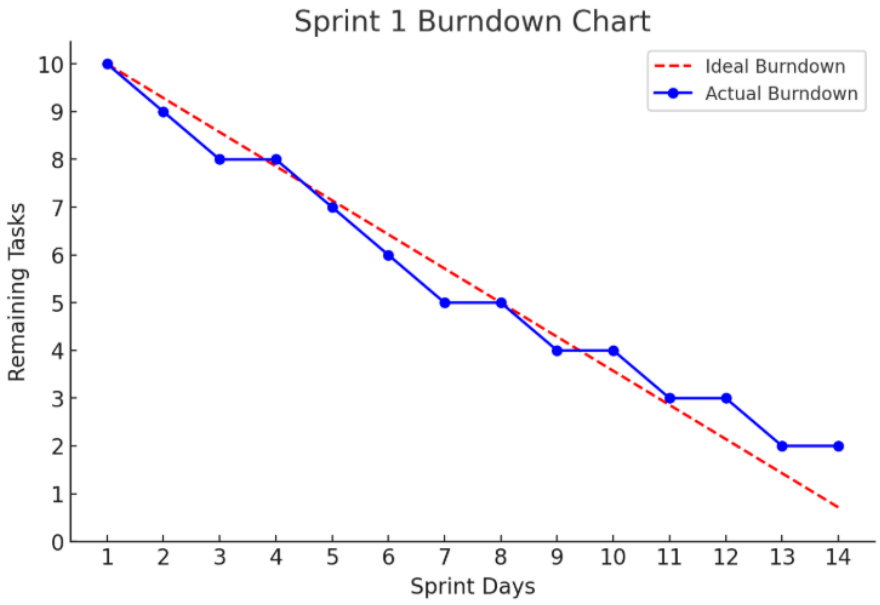
The goal is to develop a functional grocery list with basic tracking and management features to streamline the user's shopping experience.

❖ **Sprint Backlog**

- The highest-priority user stories, which were discussed during Sprint 1 and have been classified into tasks as follows:
  - ➢ Create, edit, and delete items
  - ➢ Categorize grocery items
  - ➢ Simple UI/UX design
  - ➢ Basic data storage

- The story points for each of the tasks listed above are – 1, 2, 3 and 5 respectively.
- These tasks will be implemented and tested regularly by the team in future sprints to ensure that the results align with the product backlog as well as the stakeholder feedback.

Burndown Chart:



Here's the Sprint 1 Burndown Chart!

- The red dashed line represents the ideal progress (linear completion of tasks).
- The blue line with markers shows the actual task completion, where some delays occurred mid-sprint.
- By the end of the sprint, some tasks remained incomplete, meaning which will be carried over to Sprint 2.

Sprint Review

● Sprint Review Meeting

At the culmination of Sprint 1, the team convened a Sprint Review Meeting to present the completed product increment to stakeholders, simulating a real-world demonstration to a Product Owner and customer representatives. The session was instrumental in closing the feedback loop and validating whether the sprint deliverables aligned with stakeholder expectations.

● **Present Increment**

The team showcased a functional prototype of the Smart Grocery Shopping List App, which incorporated several core capabilities committed to in the Sprint Backlog:

- Grocery List Management: Users could seamlessly add, edit, delete, and categorize grocery items.
- Smart Suggestions: The app demonstrated early-stage implementation of suggestion algorithms based on prior user interactions.
- Barcode Scanning: Basic integration of a barcode scanner allowed item entry and pantry tracking.
- Pantry Inventory: Inventory updates and manual entries were functional, laying groundwork for expiry tracking features.
- Live demonstrations illustrated how these features enhanced user convenience, efficiency, and grocery planning intelligence.

● **Stakeholder Feedback**

Stakeholders provided valuable insights during the session, emphasizing the following:

- The categorization and item-editing UI was intuitive and user-friendly.
- Suggestions and reminders were promising but would benefit from tighter AI integration and user personalization.
- Interest was expressed in enhancing the barcode database to reduce manual entry dependence.
- Real-time multi-user collaboration and meal planning features were seen as strategic differentiators and should be prioritized in future sprints.
- This feedback will directly influence the Sprint 2 backlog refinement and prioritization process.

● **Outcome**

The Product Owner evaluated the Sprint 1 outcomes and confirmed that the Sprint Goal was successfully achieved. All user stories targeted in the Sprint Backlog were completed, tested, and

integrated into the working product increment. The functional demonstration validated the core foundation for more advanced features in upcoming sprints.

The Sprint Review also served as a strategic checkpoint, enabling re-alignment of future deliverables with stakeholder expectations. It reinforced the team's agile maturity, particularly in transparent communication, adaptability, and value-driven delivery.

Challenges Faced:

Duplicate Items in Grocery Lists :-

- ♦Implement a smart item recognition system that suggests merging similar items instead of duplicating them.
- ♦Use NLP to detect similar entries.
- ♦To handle duplicate items in grocery lists, we preprocess input by converting text to lowercase, removing special characters, and standardizing units.
- ♦We then tokenize the item names and normalize them using NLP techniques like stemming.
- ♦Next, we compare new items with existing ones using smart text-matching techniques. We check for small differences in spelling like "milk whole" vs. "whole milk" and find words that mean the same thing like "soda" and "soft drink". If an item is very similar to one already on the list, we suggest merging them to avoid duplicates.
- ♦If the similarity is above 90%, we prompt the user to confirm merging the items. Once confirmed, we merge the items by updating quantities.
- ♦The system also learns from user feedback to improve future duplicate detection.

Barcode Scanner Inaccuracy:-

- ♦Fails to recognize certain products.
- ♦Allow manual entry for unrecognized items. Integrate an external product database to enhance barcode recognition.

♦If the barcode scanner fails to recognize certain products, users should have the option to manually enter the item details. Additionally, integrating an external product database, such as Open Food Facts, can improve barcode recognition by providing a larger and more accurate database of grocery items.

♦This ensures a smoother user experience and reduces the chances of missing products.

Data Loss Risk:-

♦Users accidentally delete items or lists

♦Introduce a "Recently Deleted" recovery option to restore deleted lists/items within a certain timeframe.

♦To prevent data loss from accidental deletions, a "Recently Deleted" recovery feature can be introduced, allowing users to restore deleted items or lists within a specific timeframe of 30 days.

♦When an item is deleted, it moves to a temporary trash or archive instead of being permanently removed. Users can access this section to restore mistakenly deleted entries or choose to permanently delete them.

♦ This approach ensures data safety while giving users flexibility to recover important information.

Miscommunication in Shared Lists:-

♦Family members buy the same item.

♦Implement real-time notifications when someone updates the shared list. Add an "Already Purchased" feature to notify users of recent purchases.

♦To prevent miscommunication in shared grocery lists, real-time updates and notifications are essential. Whenever a family member adds, edits, or removes an item, all other users should receive instant alerts to stay informed.

♦ A "live sync" feature ensures that changes reflect across all devices in real time. Additionally, an "Already Purchased" label can appear next to items that have been recently bought, preventing duplicate purchases.

♦This feature can work by allowing users to mark items as "Purchased" when they check out, automatically notifying others. For better tracking, a purchase history log can be added, showing who bought what and when.
♦Integrating these features enhances collaboration, avoids unnecessary spending, and improves shopping efficiency.

Accomplished So Far:

♦So far, significant progress has been made in developing the Smart Grocery Shopping List App. The core grocery list management feature has been built, allowing users to add, edit, and organize items efficiently.

♦Smart recommendations have been integrated, suggesting items based on previous purchases and meal planning habits. Additionally, a barcode scanning system has been implemented to help users quickly add items to their pantry while tracking their expiration dates to reduce food waste.

♦The initial phase of testing and bug fixes has also been completed, ensuring a smoother user experience by identifying and resolving early issues. These accomplishments lay a strong foundation for the application's functionality and user convenience.

Next Steps to be Accomplished:

♦The next steps in the development of the Smart Grocery Shopping List App focus on enhancing user experience and finalizing the product. One major improvement is the integration of AI-driven meal planning suggestions, which will analyze users' pantry inventory, past purchases, and dietary preferences to recommend personalized meal plans.

♦Additionally, the notification system will be optimized to ensure timely reminders for shopping, expiration alerts, and shared list updates. The user interface will also be refined for better usability and aesthetics.

♦As the project nears completion, a final round of testing and bug fixes will be conducted to ensure smooth functionality.

♦Lastly, a PowerPoint presentation will be prepared to showcase the application’s features, development process, and key benefits, making it ready for presenting.